

Statistical and Computational Models

Lecture 11: Catch-up on ordered, unordered and count outcomes

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Roadmap

Ordered outcomes

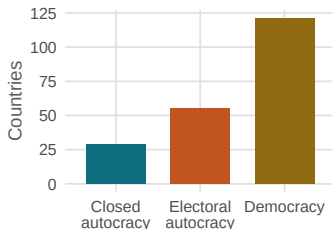
Unordered outcomes

Counts

Ordered outcomes

Regime type, $Y_i \in \{1, 2, 3\}$, is **ordinal**: it has an order, but no known spacing.

regime	y_num	lgdp	oil
Democracy	3	9.23	0
Democracy	3	8.32	0
Closed autocracy	1	6.49	1
Electoral autocracy	2	7.56	1

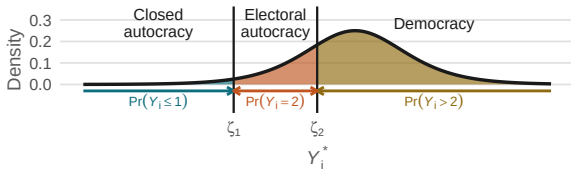


Least squares on the codes (`y_num`) assumes **equal spacing**; a logit would necessarily **group two categories** as one. We want a model that uses the order (like OLS), but assumes nothing about spacing (like logit).

Ordered logit

A latent propensity, $Y_i^* = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, logistic ε_i : one cutpoint at 0 gives the **logit**.

With two **cutpoints**, ζ_1 and ζ_2 , it is cut into the three regimes:



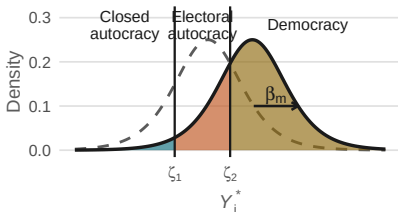
$$\mathbb{P}(Y_i \leq j \mid \mathbf{x}_i) = \text{logit}^{-1}(\zeta_j - \mathbf{x}_i^\top \boldsymbol{\beta})$$

Read: the probability of being in category j or below. By construction, a larger $\mathbf{x}_i^\top \boldsymbol{\beta}$ makes that probability **smaller**, pushing a country up the order.

Fitting ordered logit in R

```
fit_ologit <- MASS::polr(regime ~ lgdp + oil, data = countries,  
                          Hess = TRUE)
```

	Estimate	SE
lgdp	1.96	0.24
oil	-1.54	0.37
$\hat{\zeta}_1$	13.75	1.89
$\hat{\zeta}_2$	16.30	2.03



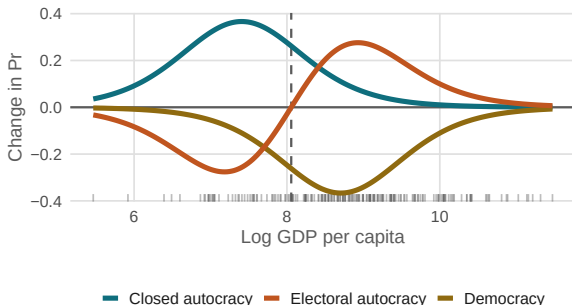
Rows 1–2 are $\hat{\beta}$; rows 3–4 the cutpoints, which absorb the intercept.

Raise x_{im} by one: the density slides right by β_m (dashed: before), so the log odds of being above **every** cutpoint rise by β_m .

Switching `oil` on multiplies those odds by $e^{\hat{\beta}_{oil}} = 0.21$.

Interpreting oil, *ceteris paribus*

Switch oil on: the change in fitted Pr, regime by regime:



Oil raises $\Pr(\text{closed})$ and lowers $\Pr(\text{democracy})$ everywhere; for electoral autocracy the direction is **ambiguous**, flipping sign at the dashed line.

Interpreting oil, *ceteris paribus*

Averaging each curve over the observed countries gives one **APE** per regime:

```
margineffects::avg_comparisons(fit_ologit, variables = "oil") |>  
  as.data.frame() |> select(group, estimate, std.error) |>  
  print(digits = 2)
```

	group	estimate	std.error
1	Closed autocracy	0.127	0.033
2	Electoral autocracy	0.091	0.022
3	Democracy	-0.218	0.050

The three APEs sum to zero. Why?

Each country's probabilities sum to one, with or without oil: whatever oil adds to one category must come out of the others.

Roadmap

Ordered outcomes

Unordered outcomes

Counts

The multinomial logit

Now three **unordered** autocracy types: military, single-party, personalist.

$$\mathbb{P}(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\mathbf{x}_i^\top \beta_j)}{\sum_{k=1}^J \exp(\mathbf{x}_i^\top \beta_k)}, \quad \beta_1 = \mathbf{0}$$

Read: category j has its own latent $Y_{ij}^* = \mathbf{x}_i^\top \beta_j + \varepsilon_{ij}$; the largest wins. Each category's **score** is $\exp(\mathbf{x}_i^\top \beta_j)$, its probability that score's **share**, with a **base category** at **0**.

Divide by the base: each β_j is a **binary logit** against the base, fit jointly.

$$\log \frac{\mathbb{P}(Y_i = j \mid \mathbf{x}_i)}{\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)} = \mathbf{x}_i^\top \beta_j$$

Fitting multinomial logit in R

```
fit_mnlogit <- nnet::multinom(autocracy ~ lgdp + oil, data = countries,  
                             trace = FALSE)  
round(coef(fit_mnlogit), 2)
```

	(Intercept)	lgdp	oil
Single-party	-12.78	1.75	-0.94
Personalist	-25.68	3.34	-2.37

Category	Coefficient on oil	APE of oil on Pr
Military	–	0.101
Single-party	-0.938	0.128
Personalist	-2.372	-0.229

Each coefficient is **relative to the base**, military. Single-party's coefficient on oil is negative, yet oil **raises** its probability.

Oil cuts personalist's score far more, and that lost share goes to the other two.

Irrelevant alternatives

Commuters split evenly between a car and a red bus. Add a blue bus, identical but for its colour (shares in %):

Option	Before	After, under IIA	After, more likely
Car	50	33	50
Red bus	50	33	25
Blue bus	–	33	25

$$\frac{\mathbb{P}(Y_i = j \mid \mathbf{x}_i)}{\mathbb{P}(Y_i = k \mid \mathbf{x}_i)} = \exp(\mathbf{x}_i^\top (\beta_j - \beta_k))$$

Read: in the multinomial logit, the ratio ignores every other category, however close a substitute: **independence of irrelevant alternatives** (IIA).

Nested logit, multinomial probit and mixed logit relax it (won't cover).

Roadmap

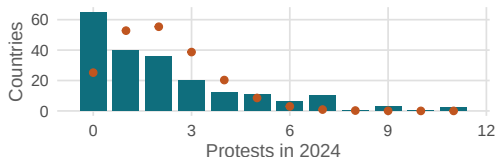
Ordered outcomes

Unordered outcomes

Counts

Counting protests

Counts, $Y_i \in \{0, 1, 2, \dots\}$: OLS can fit **negative** counts.



$$\mathbb{P}(Y_i = y) = \frac{\mu^y}{y!} e^{-\mu}$$

Bars: protests in 2024. Points: a **Poisson** with their mean, and far **too few zeros**. Its one parameter, μ , is mean and variance.

Covariates: `lgdp`, `oil`, `e_autocracy` (1 if an electoral autocracy).

Poisson regression

$$\log \mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \mathbb{E}[Y_i | \mathbf{x}_i] = \text{Var}[Y_i | \mathbf{x}_i] = \mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta})$$

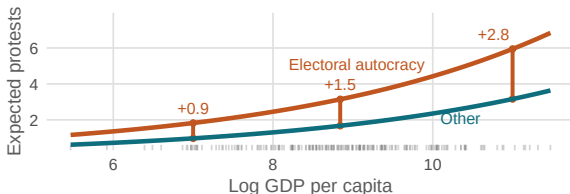
```
fit_pois <- glm(protests ~ lgdp + e_autocracy + oil, data = countries,  
               family = poisson)  
round(coef(summary(fit_pois))[, 1:2], 2)
```

	Estimate	Std. Error
(Intercept)	-2.10	0.46
lgdp	0.30	0.05
e_autocracy	0.63	0.11
oil	-0.03	0.11

The **log** link keeps μ_i positive: a count needs only a floor.

$e^{\hat{\beta}_m}$ is a **ratio of expected counts**: electoral autocracies expect 1.88 times as many protests, at the same `lgdp` and `oil`.

Interpreting regime type, *ceteris paribus*



Expected protests at $\text{oil} = 0$. The ratio of the curves is 1.88 at every lgdp ; the **gap** is not.

```
marginaleffects::avg_comparisons(fit_pois, variables = "e_autocracy") |>  
  as.data.frame() |> select(estimate, std.error) |>  
  print(digits = 2)
```

	estimate	std.error
1	1.5	0.31

Read: about 1.5 more protests a year than otherwise similar countries, on average.

Lecture 10: "Interpreting regime type, *ceteris paribus*" (both frames).

The variance is usually wrong

For many counts, $\text{Var}[Y_i | \mathbf{x}_i] > \mathbb{E}[Y_i | \mathbf{x}_i]$: **overdispersion**, from unobserved heterogeneity or contagion.

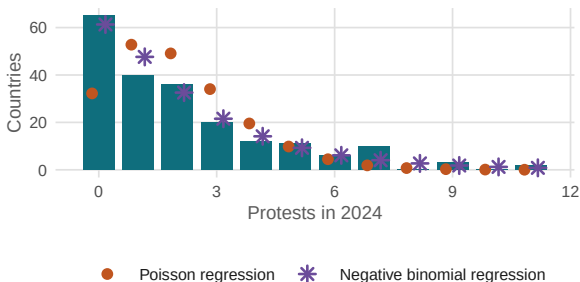
$\hat{\beta}$ is still **consistent**: it needs only the mean to be right. But the Poisson's standard errors are **too small**.

	Estimate	Poisson se	Quasi-Poisson se	Sandwich se
(Intercept)	-2.098	0.458	0.708	0.674
lgdp	0.295	0.048	0.075	0.070
e_autocracy	0.632	0.110	0.170	0.173
oil	-0.035	0.110	0.171	0.166

Quasi-Poisson ($\hat{\phi} = 2.39$) and the sandwich give the same $\hat{\beta}$, but more conservative standard errors.

Negative binomial

Rather than rescuing the Poisson regression, try a different **probability distribution**: one whose variance can grow faster than its mean.



Points: the expected number of countries at each count, under each fitted regression.

Negative binomial

The **negative binomial** adds one parameter, θ :

$$\mathbb{E}[Y_i | \mathbf{x}_i] = \mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}), \quad \text{Var}[Y_i | \mathbf{x}_i] = \mu_i + \mu_i^2 / \theta$$

Read: the **same** mean as the Poisson; μ_i^2 / θ lets the variance outrun it. As $\theta \rightarrow \infty$, it becomes the Poisson.

MASS: : `glm.nb()`: $\hat{\theta} = 1.42$. It uses the log link, so $\mathbf{e}^{\hat{\beta}_m}$ reads as in Poisson regression.